JC2001 Introduction to Software Engineering

Practical Session 3 - Exercises

Software Project Plan

The Software Development Life Cycle (SDLC), also called the Application Development Life Cycle, is a structured process that enables the production of high-quality, low-cost software in the shortest possible time. The SDLC is a process for planning, creating, testing and deploying an information system. There are usually seven stages in the SDLC: (1) system investigation, (2) requirements analysis, (3) design, (4) development and testing, (5) implementation, (6) documentation and (7) evaluation.

In the 1950s and 1960s, computer science progressed rapidly. This shift in evolution sparked the beginnings of a production framework that eventually grew into the SDLC we know today. Before the 1950s, computing was not complex enough to require a detailed approach like the SDLC. As the complexity and scale of programming grew, the concept of structured programming emerged. Over time, structured programming demanded more tactical development models, thus sparking the beginnings of the SDLC.

The system investigation stage, also known as the planning stage, is when the project plan is developed to identify, prioritise, and assign the tasks and resources required to build the project structure. It helps define the problem and scope of any existing systems, as well as determine the objectives for the new system. Most importantly, the planning stage sets the project schedule and culminates in a detailed project plan. In this exercise, we will develop the tasks and deliverables and put them into a detailed project plan.

Exercise A: Tasks and Deliverables for a POS System

The retail industry is a major user of POS terminals. A retail POS system typically includes a cash register and usually includes a debit/credit card reader. It may also include a conveyor belt, checkout divider, weight scale, an integrated debit and credit card processing system, a signature capture device and a customer PIN pad. The POS unit handles sales to the consumer, but it is only one part of the entire POS system used in a retail business. Back-office computers typically handle other POS functions, such as inventory control, purchasing, receiving, and transferring products to and from other locations. Other typical POS functions include storing store sales information to enable customer returns, reporting, sales trends, and cost-price-profit analysis. Customer information may be stored for receivables management, marketing purposes and specific buying analysis. Many retail POS systems include an accounting interface that feeds sales and cost-of-goods information to independent accounting applications.

Question 1: List all the tasks required for developing a POS system.

Question 2: List the deliverables for the POS system.

Exercise B: Project Plan

Once you have listed all the tasks and deliverables, you will develop a detailed project plan. A project plan, also known as the project management plan, is the document that describes how the project will be executed, monitored, controlled and closed. It outlines the project's objectives and scope and serves as an official reference for the project team, the larger company, and stakeholders.

Project planning is important because it provides structure and foresight for the execution stage, helping to eliminate wasteful activities and patterns.

Question: Develop a project management plan for the design and development of a POS system for a local supermarket. In your project plan, include all stages of the waterfall model and all required tasks and deliverables.

References

1. What is SDLC? https://www.synopsys.com/glossary/what-is-sdlc.html
2. System Development Life Cycle, https://www.clouddefense.ai/blog/system-development-life-cycle
3. SDLC in Project Management, https://pmbasics101.com/software-development-life-cycle/